

Toxic cocktails in our soils : Assessing fungicide-insecticide mixture effects on earthworm growth and reproduction

Supplementary information

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1 Earthworm culture

1.1 Breeding and experiment soil

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay ($< 2 \mu\text{m}$, g/kg)	226.0
Fine silt (2-20 μm , g/kg)	174.1
Coarse silt (20-50 μm , g/kg)	298.9
Fine sand (50-200 μm , g/kg)	239.1
Coarse sand (200-2000 μm , g/kg)	47.9
CaCO_3 total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

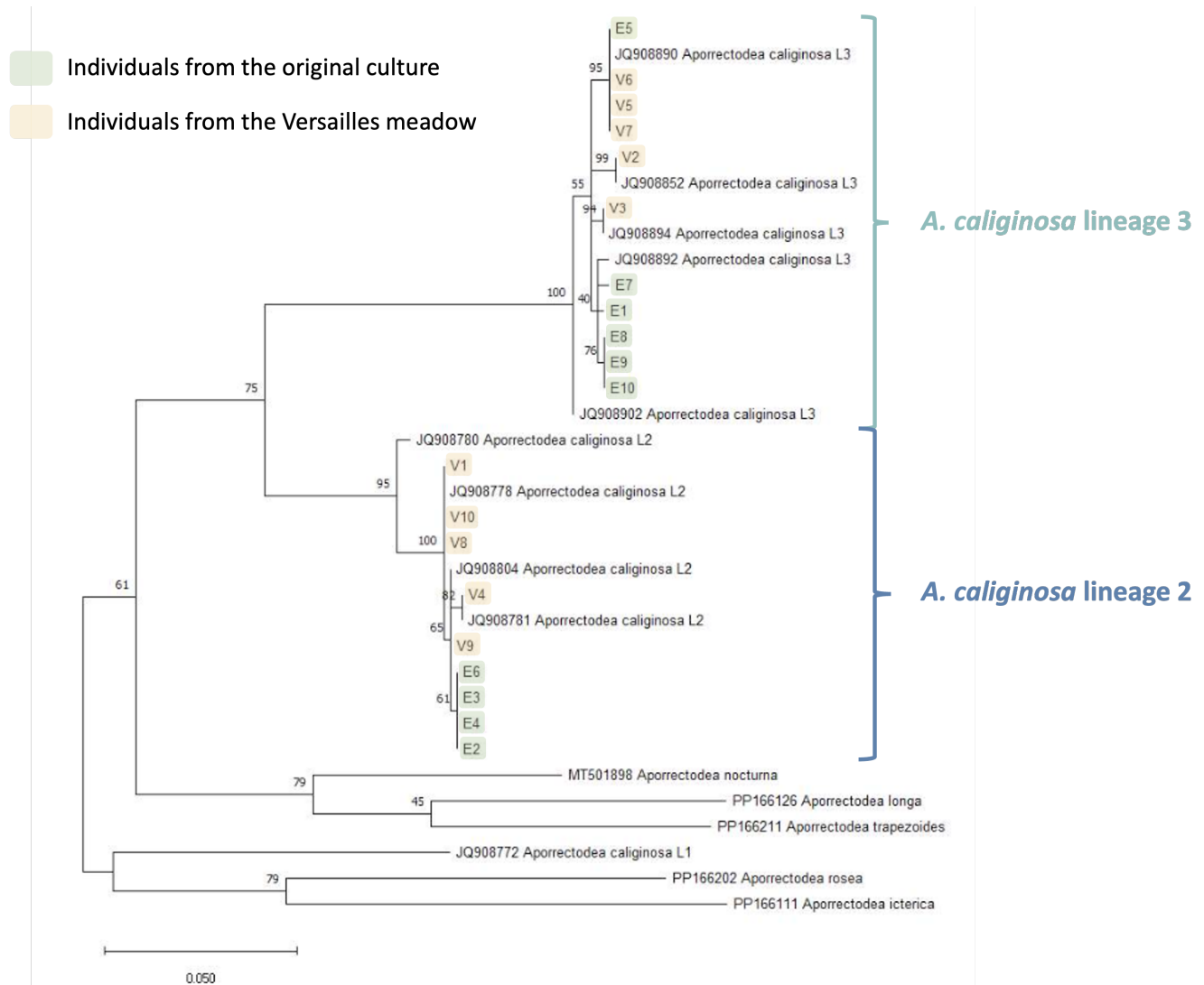


Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

2 Experiments

2.1 Growth and reproduction experiments



Figure 2.1: Overview of the growth experiments

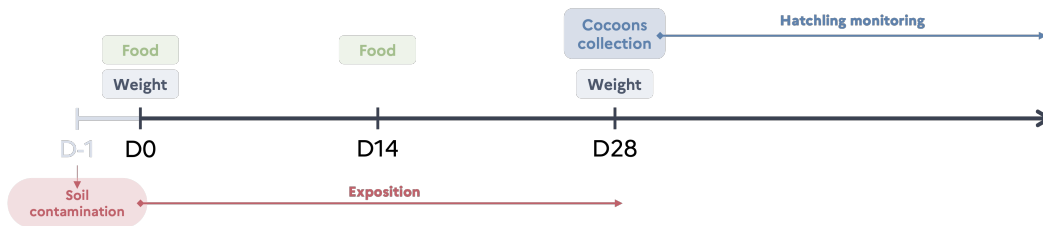


Figure 2.2: Overview of the reproduction experiments

2.2 Experiment A - Single substance

2.2.1 Juvenile growth

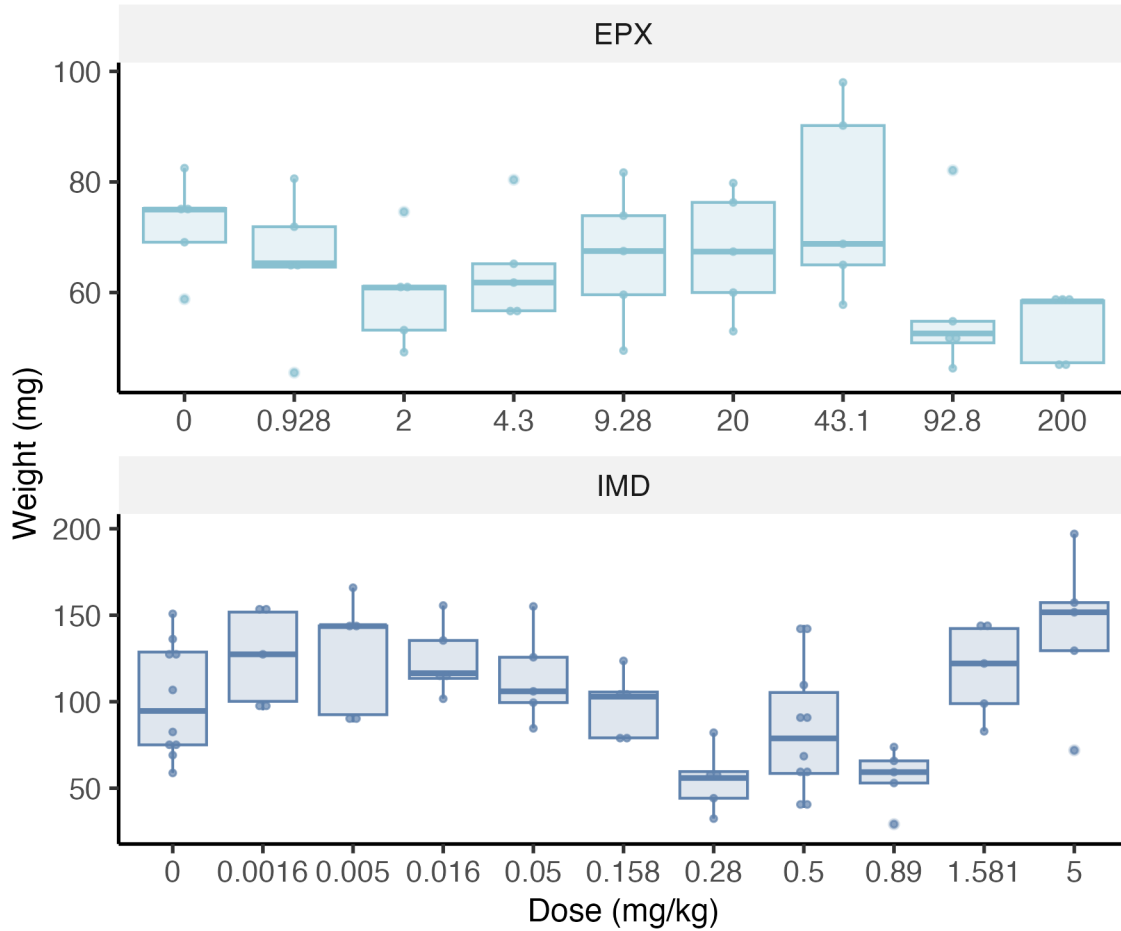


Figure 2.3: Initial weight of juvenile earthworms.

For EPX, there is a significant effect of the dose on earthworm initial weights (Student test : $pval(EPX) = 0.028$). However, earthworms were non the less randomly attributed to conditions and the growth model allows to take the differences in initial weight into account. For IMD, there is no significant effect of the dose on the initial weight of earthworms (Student test : $pval(IMD) = 0.067$).

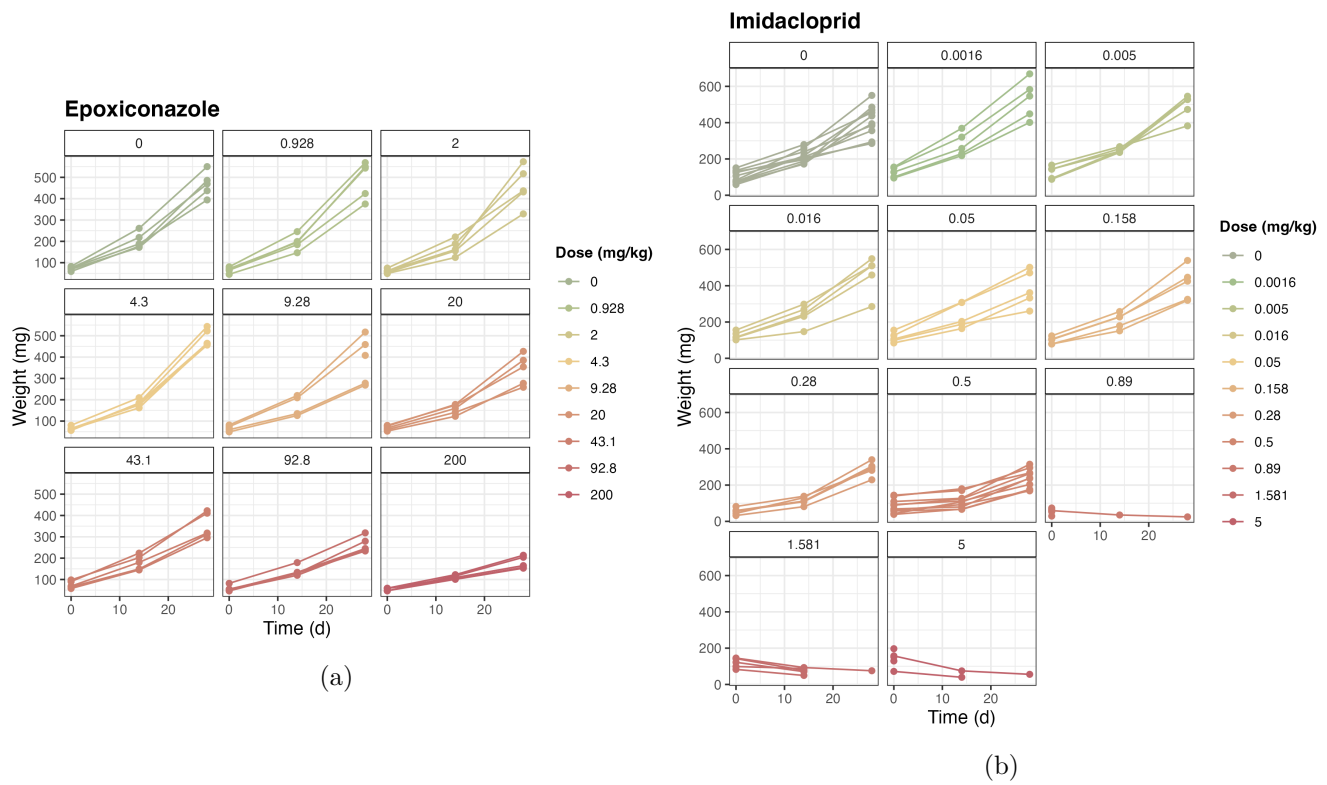


Figure 2.4: Growth curves of juvenile earthworms.

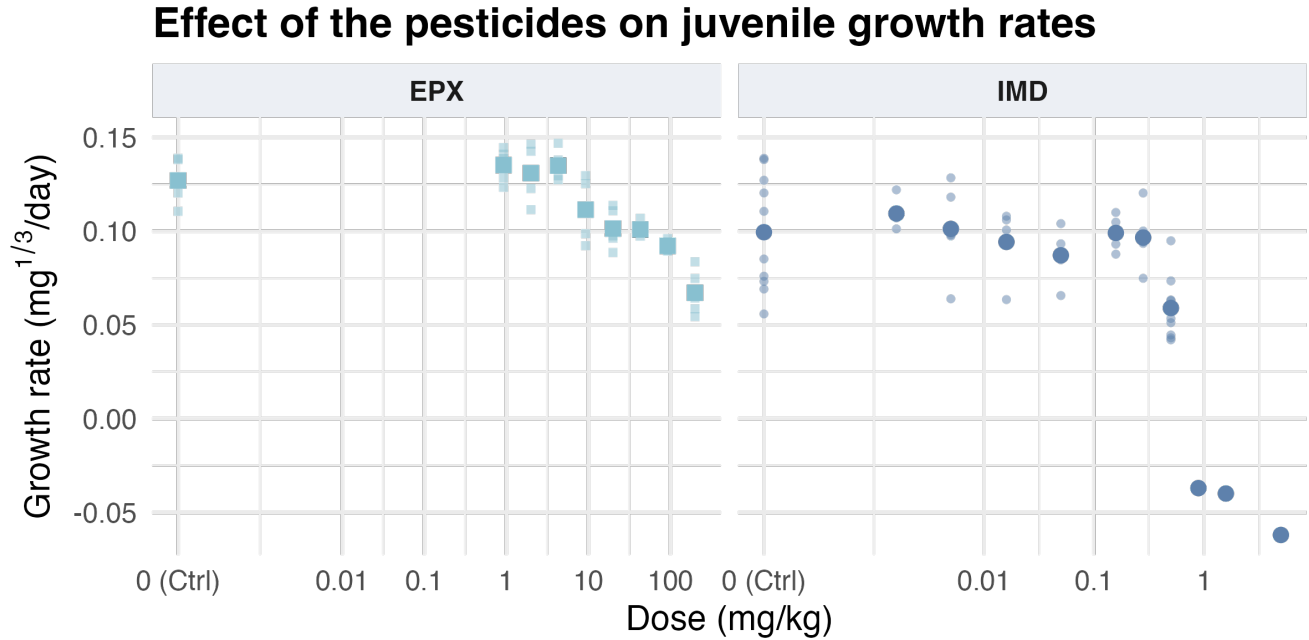


Figure 2.5: Effect of the pesticides on juvenile growth rates

Table 2.1: NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method.

	NOEC (mg/kg)	LOEC (mg/kg)
IMD	0.28	0.5
EPX	9.28	20

2.2.2 Reproduction

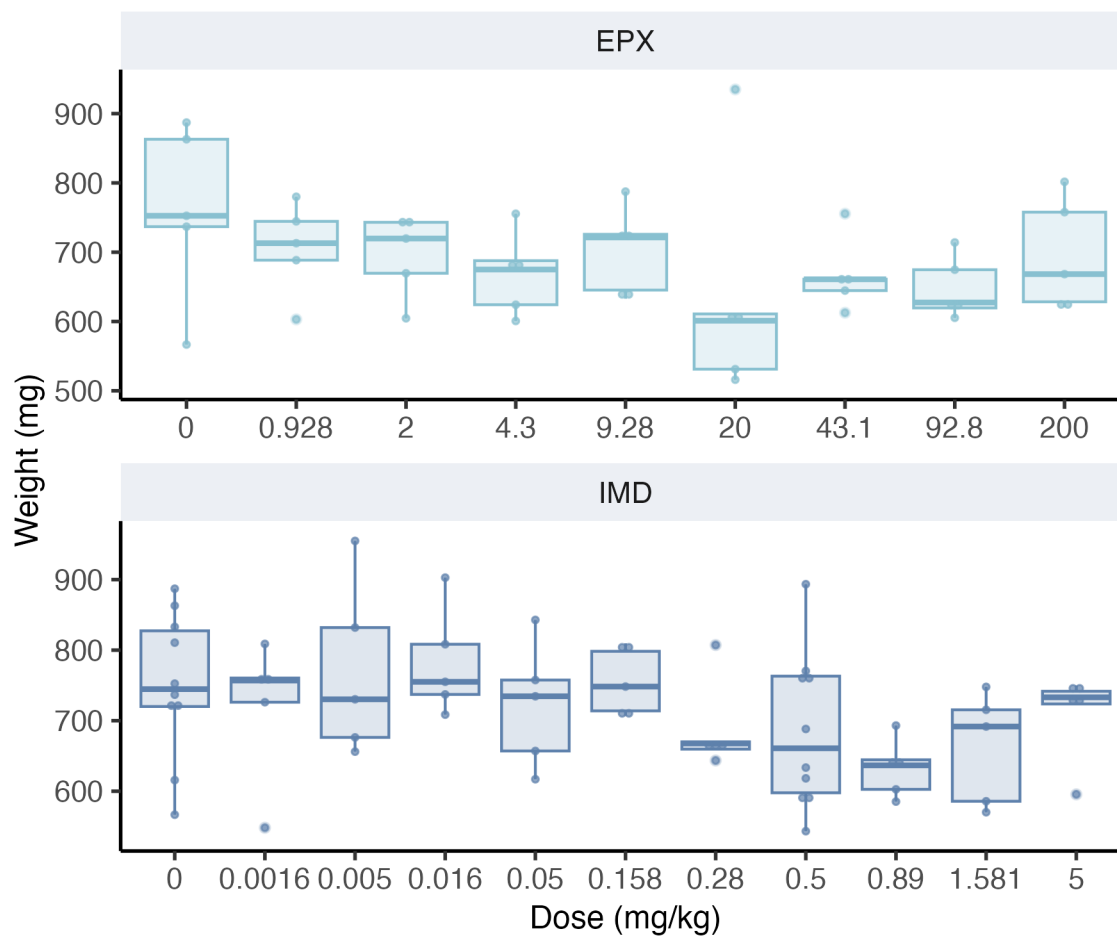


Figure 2.6: Initial weight of adult earthworms.

There is no effect of the dose on the initial weight of earthworms (Student test : $pval(EPX) = 0.63$ & $pval(IMD) = 0.25$).

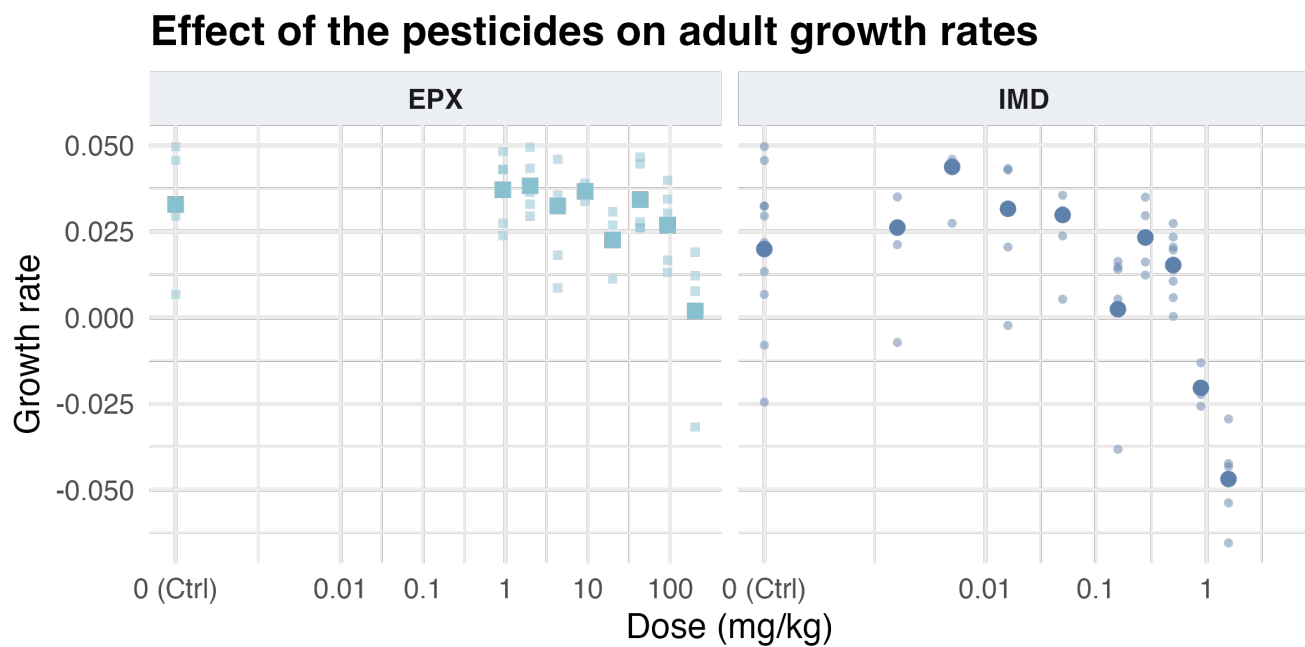


Figure 2.7: Effect of the pesticides on adult growth rates.

Table 2.2: NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method.

	NOEC (mg/kg)	LOEC (mg/kg)
IMD	0.5	0.89
EPX	92.8	200

2.3 Experiment B - Mixture

3 Jonker interaction models

3.1 Mathematical formulation

Let f_A and f_B be the functions describing the dose-response curves of substance A and substance B, respectively.

In the case of no interaction, concentration addition implies the dose-response surface of the mixture of A and B can be expressed as all couple of concentration of substance A and substance B ($c_A ; c_B$) as :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} = 1 \quad (3.1)$$

The Jonker simple interaction model add the definition of an interaction parameter (a) so the dose-response surface of the mixture of A and B can be expressed as all ($c_A ; c_B$) as :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} = \exp(G) \quad (3.2)$$

With :

$$G(z_A, z_B) = a \times (z_A \times z_B) \quad (3.3)$$

And :

$$z_i = \frac{TU_{x_i}}{TU_{x_A} + TU_{x_B}}, \quad \text{where } TU_{x_i} = \frac{c_i}{EC_{x_i}} \quad (3.4)$$

With :

- Y : The response for a given couple of concentrations,
- x : Usually 5, 10, ou 50 (50 used in this study),
- z_A & z_B : Contribution of each substance to the overall effect,

For the SA model, the interaction parameter value describes the type of the interaction :

- $a = 0$: Additivity,
- $a < 0$: Synergy,

- $a > 0$: Antagonisme.

The Jonker framework adds two other types of model, making it four models in total :

- CA : Concentration addition, case with no interaction (Equation 3.1)
- SA : Simple interaction with constant deviation from additivity (Equation 3.3),
- DL : Dose-level dependent interaction where effects vary with response magnitude (Equation 3.5),
- DR : Ratio-dependent interaction where effects vary with mixture composition (Equation 3.5).

The DL and DR models include a second interaction parameter, b . The expression of G is then :

$$G(z_A z_B) = \begin{cases} a[1 - b_{DL}(TU_{50_A} + TU_{50_B})] z_A z_B & \text{(DL model)} \\ (a + b_{DR} z_A) z_A z_B & \text{(DR model)} \end{cases} \quad (3.5)$$

3.2 Computational implementation

In practice, the fit of the dose-response curve surface is a double optimization :

1. For given set of interaction parameters, the corresponding effects must be found also numerically.
2. Numerical optimization of the interaction parameters

To perform step 1, a function `CA_complete2()` was written. For a given set of interaction parameters, finding the corresponding effect for a couple of concentration (c_A ; c_B) implies the minimization of the following term :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} - \exp(G) \quad (3.6)$$

For step 2, a function `CA_complete_fit_speed()` was written. Parameter estimation for the interaction models requires maximum likelihood estimation (MLE). The optimization procedure depends on the assumed error distribution: for normally distributed errors, this is equivalent to minimizing the residual sum of squares (calculated with the function `CA_complete2_RSS()`), while for Poisson-distributed errors, the log-likelihood function (Equation 3.7) (calculated with the function `CA_complete2_Poisson()`) is directly maximized.

Let $\mathbf{y} = (y_1, y_2, \dots, y_n)$ be the vector of observed responses, the parameter vector to be estimated, and $f(.|\cdot)$ the density function conditional to the parameters. Then, the likelihood function is defined as :

$$\mathcal{L}(\theta|\mathbf{y}) = \prod_{i=1}^n f(y_i|\theta)$$

The log-likelihood function, which is computationally more tractable, is given by:

$$l(\theta|\mathbf{y}) = \ln(\mathcal{L}(\theta|\mathbf{y})) = \sum_{i=1}^n \ln(f(y_i|\theta))$$

For count data (cocoen production), assuming $y_i \sim \mathcal{P}(\lambda_i)$ where $\lambda_i = g(\mathbf{x}_i; \theta)$ represents the expected response given covariates $\mathbf{x}_i$, the log-likelihood becomes:

$$\begin{aligned} l(\theta|\mathbf{y}) &= \sum_{i=1}^n \ln \left(\frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!} \right) \\ &= \sum_{i=1}^n [y_i \ln(\lambda_i) - \lambda_i - \ln(y_i!)] \\ &= \sum_{i=1}^n [y_i \ln(g(x_i; \theta)) - g(x_i; \theta) - \ln(y_i!)] \end{aligned} \tag{3.7}$$

References

Bart, S., Laurent, C., Péry, A. R. R., Mougin, C., & Pelosi, C. (2017). Differences in sensitivity between earthworms and enchytraeids exposed to two commercial fungicides. *Ecotoxicology and Environmental Safety*, 140, 177–184. <https://doi.org/10.1016/j.ecoenv.2017.02.052>